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MILITARY SPECIFICATION

BEEF TIPS WITH GRAVY, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.. This document covers beef tips with gravy, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-495 - Coatings, Exterior, for Tinned Food Cans
PPP-B-636 - Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Personal Equipment and Equipage; General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Institutional Meat Purchase Specifications for Beef

(Copies may be obtained from the Director, Livestock and Seed Division, Agricultural Marketing Service, P.O. Box 96456, U.S. Department of Agriculture, Room 2092, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

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Meat and Poultry Inspection Regulations

Food Safety and Inspection Services, Canning of Meat and Poultry Products (9 CFR Part 318)

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. Standards for Grades of Tomato Paste

(Copies may be obtained from the Director, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. Standards for Condition of Food Containers

(Copies may be obtained from the Chairman, Condition of Food Container Committee, Agricultural Marketing Service, Room 0610, South Building, P.O. Box 96456, U.S. Department of Agriculture, Washington, DC 20090-6456.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies may be obtained from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Copies should be obtained from the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

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ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Copies should be obtained from the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Copies should be obtained from the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Copies should be obtained from the American Dehydrated Onion and Garlic Association, 650 California Street, Suite 800, San Francisco, CA 94108.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-sensitive Tape

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

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3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any combination or portion of recognizable primal rounds (shank off, heel out, and knuckles, peeled) or subprimal cuts of the round, shank off (top round, bottom round, heel out, and knuckles, peeled). Recognizable cuts are those which, when compared to Institutional Meat Purchase Specifications (IMPS) cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. A portion is a certified recognizable cut from which objectionable tissue, such as tendons or heavy connective tissue, has been removed. The beef shall be in the fresh-chilled state and shall be in excellent condition, i.e. exposed lean and fat surfaces shall be of a color and bloom normally associated with the class and cut of meat and typical of meat which has been properly stored and handled. Cut surfaces and naturally exposed lean surfaces shall show no more than slight darkening or discoloration due to dehydration, aging, or microbial activity. The fat shall show no more than slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum-packaged meat in excellent condition shall be acceptable. Also, meat shall show no evidence of freezing, defrosting, or mishandling.

3.2.1.1 Boning and trimming. The beef shall be boned and trimmed to remove objectionable material such as bone, cartilage, or heavy connective tissue. The boneless trimmed beef shall meet the limitations specified in tables I and II.

3.2.1.2 Handling and storage. Handling and storage of the boned and trimmed beef prior to processing into the finished product shall be in accordance with the following requirements:

a. Beef processed on the day of initial certification shall be maintained in the temperature range of 28° to 50°F (inclusive).

b. Holding in the fresh chilled state for not more than 4 days after initial certification is permitted provided that the beef is maintained in the temperature range of 28° to 40°F (inclusive).

c. Holding in the frozen state for not more than 120 days after initial certification is permitted provided that the beef is:

- frozen to 0°F or lower within 72 hours after initial certification
- stored at 0°F or lower
- protected from freezer deterioration and damage
- stored in containers that are adequate to maintain product excellence
- held after storage at an internal temperature not to exceed 40°F when further processing is resumed.

3.2.2 Water. Water used for formulation, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

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3.2.3 Mushrooms, frozen, sliced. The frozen mushrooms shall be prepared from fresh, sound, properly trimmed and washed mushrooms conforming to the characteristics of the species Agaricus (pasalliota) bisporous or Agaricus bitorguis. The mushrooms shall be of the white or cream color type and shall be of a randomly sliced style. The frozen, sliced mushrooms shall be of the latest date of pack.

3.2.4 Starch, food, modified, high opacity. The high opacity starch shall be a white, odorless, finely pulverized, modified waxy maize starch for use in thermostabilized foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The cooked starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

3.2.5 Vegetable oil. The vegetable oil shall have a stability of 15 hours (minimum), have a free fatty acid content not to exceed 0.05 percent by weight, have a moisture/volatile matter not to exceed 0.06 percent by weight, and pass the cold test of 5.5 hours (minimum).

3.2.6 Tomato paste. The tomato paste shall be U.S. Grade A or better of the U.S. Standards for Grades of Tomato Paste (see 3.3.3 for soluble solids content).

3.2.7 Soup stock, dehydrated. The dehydrated soup stock shall be a light brown to tan, free flowing powder derived from liquid soup stock. The soup stock shall be made from USDA inspected beef bones, or pork bones, with adhering meat or a combination of the two materials and shall have a minimum of 93 percent total solids and a maximum of 2 percent fat.

3.2.8 Onion powder. Onion powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.9 Salt. Salt shall be noniodized, clear, white, refined, sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.10 Sugar, white, granulated. Sugar shall be white, refined, sucrose, granulated cane or beet or a combination thereof and shall comply with MIL-STD-900.

3.2.11 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.12 Pepper, black, ground. The ground black pepper shall have been ground from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground pepper shall have a characteristic, penetrating odor, a hot biting pungent flavor, and a light gray to speckled black-gray

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color. The black pepper shall contain not less than 2.0 mL of volatile oil per 100 grams and be of such size that 95 percent shall pass through a U.S. Standard No. 16 sieve.

3.2.13 Celery seed, ground. Ground celery seed shall be prepared from the seed of Apium graveolens L., shall be light to rich brown in color and possess a characteristic celery odor and flavor with a warm to slightly bitter after taste. Volatile oil content shall be not less than 2.0 mL/100g and the celery seed shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 55 sieve.

3.2.14 Garlic powder. Garlic powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.15 Bayleaves, ground. Ground bayleaves shall be derived from the dried leaves of Laurus nobilis L. It shall possess a pleasant aromatic odor, a pungent, mildly bitter flavor, and a pale green to yellow green color. Volatile oil content shall be not less than 1.0 mL/100g and the bayleaves shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 30 sieve.

3.2.16 Allspice, ground. Ground allspice shall be derived from dried, nearly ripe fruit of Pimenta officinalis Lindl, shall have a fragrant, clovelike aroma, strongly aromatic, pungent, clovelike flavor, and a dark, reddish brown color. The volatile oil content shall be not less than 3.0 mL/100g and the allspice shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 25 sieve.

3.2.17 Nutmeg, ground. Ground nutmeg shall be derived from the dried seeds of Myristica fragrans Houtt, and shall have a dull brown color. The volatile oil content shall be not less than 7.5 mL/100g and the nutmeg shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 20 sieve.

3.2.18 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and further processing. Processing shall be on a continuous basis.

3.3.1 Beef preparation. The beef shall be prepared as follows:

a. Boned and trimmed meat meeting the requirements of 3.2.1.1 shall be further trimmed, if necessary, so as to ensure compliance with finished product requirements (see 3.6).

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b. The beef shall be cut into rectangular chunks measuring approximately 1 inch by 1 inch by 1-1/2 inches. The cut beef chunks shall be blanched (see 6.4). Retain broth and rendered fat for inclusion in the gravy. The broth and rendered fat shall be incorporated in the gravy formulation immediately following blanching of the beef and used in the maximum amount available.

c. The cooked beef shall be maintained in the temperature range of 28° to 40°F for not more than 48 hours from time of cooking to time of filling into tray pack cans. Cooling is not necessary if filling occurs immediately after cooking.

3.3.2 Preparation of frozen mushrooms. The frozen mushrooms shall be reduced (by mechanical means) to a pasty consistency prior to mixing with the gravy.

3.3.3 Preparation of the gravy. The gravy shall be formulated and prepared as follows:

Ingredients	Percent by weight
Broth rendered fat, and water	69.00
Mushrooms	8.50
Starch, modified, high opacity	5.25
Vegetable oil	5.00
Tomato paste (26 percent natural tomato soluble solids) 1/	4.00
Soup stock, dehydrated	3.50
Onion powder	2.50
Salt 2/	1.25
Sugar	0.50
Lecithin	0.25
Pepper, black, ground	0.12
Celery seed, ground	0.05
Garlic powder	0.03
Bayleaves, ground	0.02
Allspice, ground	0.02
Nutmeg, ground	0.01

1/ If the specific tomato solids are not available, the tomato solids which are used must be calculated to bring the solid level to that specified.

2/ The total amount of salt in the gravy formula shall be adjusted as necessary to produce a product that complies with the finished product salt requirements (see 3.6).

a. A slurry shall be made by mixing the broth, starch, vegetable oil and part of the water.

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b. An emulsion shall be made by mixing the remainder of ingredients (except the slurry). The emulsion shall be heated to a boil in a steam jacketed kettle with continuous, vigorous agitation to attain maximum emulsification of the rendered fat.

c. The steam supply to the kettle shall be shut off and the slurry added and uniformly mixed to form the gravy.

d. The volume of the final mixture shall be adjusted with water to compensate for evaporation loss during heating and holding.

e. The gravy shall be used within 4 hours after cooking. The prepared gravy shall be held in the temperature range of 150°F to 180°F at all times prior to filling into the tray pack cans.

3.3.4 Finished product preparation. The blanched beef and gravy shall be combined in the following proportions using a two-stage filling operation:

Ingredient	Percent by weight
Blanched beef	62.50
Gravy	37.50

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product such as to conform to the finished product requirements and to the following requirements:

a. The temperature of the beef at time of filling shall be in the temperature range of 28° to 40°F unless filling of cans occurs immediately after cooking.

b. The temperature of the gravy at time of filling shall be in the temperature range of 150° to 180°F.

c. Each can shall be hermetically sealed under a vacuum established by a processing authority and specified in the scheduled process so as to assure compliance with the finished product requirements (see 3.6s).

d. Each filled and sealed tray pack can shall be in the retort process within 2 hours after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

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a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, rancid, scorched, burnt, stale, sour, rancid, moldy, or sulfurous.

c. There shall be no color foreign to the product.

d. No individual can shall contain less than 50.0 ounces drained weight of beef.

e. The average drained weight of beef shall be not less than 54.0 ounces.

f. At least 50 percent of the beef (by weight) shall be pieces measuring not less than 1/2 inch nor more than 1-1/2 inches in any dimension.

g. Surface fat on each beef chunk shall measure not more than 1/8 inch in thickness.

h. Total weight of cartilage, coarse connective tissue, sections of tendons or ligaments, and glandular material, collectively, in any individual can shall not be greater than 2 ounces.

i. No individual can shall contain a bone piece measuring 0.3 inch or more in any dimension.

j. The texture of the beef chunks shall not be dry, tough, rubbery, or mushy.

k. Gravy shall not be lumpy.

l. The average net weight shall be not less than 106 ounces.

m. No individual can shall have a net weight of less than 104 ounces of product.

n. The average fat content of the finished product shall be not greater than 7.0 percent.

o. No individual tray pack shall have a fat content greater than 9.0 percent.

p. No individual tray pack shall have a salt content greater than 1.3 percent or less than 0.5 percent.

q. The product shall show no evidence of excessive heating (materially darkened or scorched).

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r. The viscosity of the gravy shall be not less than 10.5 cm per 10 seconds nor greater than 23.0 cm per 10 seconds when determined by a Bostwick Consistometer, or not less than 1,300 centipoise nor greater than 3,550 centipoise when determined by a Brookfield Viscosimeter equipped with a No. 3 spindle stirring at a speed of 10.0 rpm (see 4.5.3.4).

s. Filled, sealed, and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

3.6.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Beef examination for condition and cut. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements in 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts or portions thereof. Any nonconforming beef shall be rejected.

4.5.1.2 Boned and trimmed beef examination. After boning and trimming and prior to any further processing or to any freezing, the beef shall be examined for the defects listed in table II. The lot size, expressed in terms of pounds, shall be declared to the Agricultural Marketing Service (AMS) agent by the contractor. However, the AMS agent reserves the right to declare as a lot a portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any requirement. The sample unit shall be a minimum of 12 pounds of adjacent boneless beef. The sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the beef to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for beef rejected because of freezing, defrosting, or not being in excellent condition, the beef may be reworked by the contractor and reoffered for examination. For reexamination, the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Beef shall not be reexamined more than one time.

TABLE I. Sampling plan for boned and trimmed beef

Lot size (pounds)	Sample size (sample units)	Defect categories			
		Major		Minor	
		AC	RE	AC	RE
500 or less	20	2	3	5	6
501 to 1,200	32	3	4	7	8
1,201 to 3,200	50	5	6	10	11
3,201 to 10,000	80	7	8	14	15
10,001 to 38,000	125	10	11	21	22
Reinspection lots of 10,001 to 38,000	200	14	15	21	22

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TABLE II. Boned and trimmed beef defects 1/ 2/

Category		Defect
Major	Minor	
101		Presence of popliteal, prescapular, prefemoral or any exposed lymph gland measuring 0.5 inch or more in any dimension.
102		Presence of blood clot measuring 1.0 inch or more in any dimension.
	201	Presence of blood clot measuring 0.5 inch or more but less than 1.0 inch in any dimension.
103		Presence of bruise measuring 1.0 inch or more in any dimension.
	202	Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension.
104		Presence of bone measuring 0.3 inch or more in any dimension.
105		Presence of cartilage measuring 0.5 inch or more in any dimension.
106		Presence of backstrap measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	203	Presence of backstrap measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
107		Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 2.0 square inches or more.

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TABLE II. Boned and trimmed beef defects 1/ 2/

Category		Defect
Major	Minor	
	204	Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or gracilis membrane) measuring 0.5 square inch or more but less than 2.0 square inches.
	205	Presence of heavy connective tissue on lower edge of short plate or flank.
	206	Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension.
	207	Presence of membranous portion of diaphragm or membranous covering from skirt, flank, or abdominal section of short plate measuring 3.0 square inches or more.
108		Presence of knuckle cover.
109		Presence of kidney, pizzle eye, prepubic tendon, thymus gland, or hanging tender measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	208	Presence of kidney, pizzle eye, prepubic tendon, thymus gland, or hanging tender measuring less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other).
	209	Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more.
	210	Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface.
	211	Presence of dehydrated surface measuring 1.0 square inch or more.

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TABLE II. Boned and trimmed beef defects 1/ 2/ - Continued

Category		Defect
Major	Minor	
	212	Presence of discolored meat (including blood discolored neck meat) measuring 1.0 square inch or more.
	213	Presence of exposed blood vessel measuring 1.0 inch or more in any dimension.
	214	Presence of cod, udder, kidney, or pelvic fat.
	215	Protruding ligaments and tendons not cut flush to the surface of the lean.

1/ Determination of wholesomeness and acceptability of product with respect to the presence of foreign material (e.g., glass, dirt, insect parts, hair, wood, metal) shall be made by a Meat and Poultry Inspection Operations employee.

2/ Evidence of freezing or defrosting or product not in excellent condition shall cause rejection of the lot.

4.5.1.3 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

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4.5.3 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

4.5.3.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table III. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be heated for 35 minutes in boiling water, opened, and inspected for the defects listed in table IV.

TABLE III. Double sampling plan for product inspection 1/

Lot size (cans)	Sample size (cans)	Cumulative sample	Acceptance number	Rejection number
0 to 3,200	8	-	0	2
	8	16	1	2
3,201 to 35,000	13	-	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and, if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

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TABLE IV. Product defects 1/ 2/

Category	Defect
Major	
101	Drained weight of beef in a can is less than 50.0 ounces. 3/ 4/
102	More than 50 percent of the meat chunks measure less than 1/2 inch or more than 1 1/2 inch in any dimension.
103	Surface fat on beef chunks measuring more than 1/8 inch in thickness.
104	Total weight of cartilage, opaque connective tissue, sections of tendons or ligaments, and glandular material, collectively, in a can is greater than 2 ounces.
105	Any individual can contains a bone piece measuring 0.3 inch or more in any dimension.
106	Beef chunks dry, tough, rubbery, or mushy. 5/
107	Gravy is lumpy.
108	Product shows evidence of excessive heating (materially darkened or scorched).

1/ The presence of foreign material (e.g., glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

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3/ To determine drained weight, the free liquid in the can shall be poured off, strained through a U.S. Standard No. 8 sieve, and reserved for viscosity determinations. The remaining contents shall be poured into a flat bottom container. A minimum of three times the tray pack can's volume of 190° to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated so as to liquify rendered fat and remove the gravy without breaking the beef pieces. The contents shall then be poured into a U.S. Standard 1/4-inch sieve in a manner that will distribute the product over the sieve without breaking the beef pieces. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at an approximate 45 degree angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

4/ Failure to meet average drained weight will result in rejection of the lot.

5/ Examination for conformance to the beef texture requirement shall be made of representative beef chunks immediately following the drained weight determinations.

4.5.3.3 Fat and salt content testing. Nine tray pack cans shall be selected at random from the lot and distributed as follows:

- Three for laboratory analysis.
- Three for submission to the contractor.
- Three for retention by the AMS agent as reserve samples.

The three tray packs for laboratory analyses shall be individually tested for fat and salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, chapter: Meat and Meat Products, Section: Meat, except that preparation of the samples shall be as follows. The three unopened cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The test results shall be reported to the nearest 0.1 percent. Any test failure is a major defect. The lot shall be rejected if any of the following major defects occur:

a. The average fat content of the three tray packs is greater than 7.0 percent.

b. The fat content of any individual tray pack is greater than 9.0 percent.

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c. The salt content of any individual tray pack is greater than 1.3 percent or less than 0.5 percent.

Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the fat or salt content requirement. When the reserve samples are analyzed, the analyses for both fat and salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.5.3.4 Viscosity testing. The strained free liquid collected from each of the cans in the first sample of cans selected for drained weight inspection (see 4.5.3.2 and footnote 3/ to table IV) shall be individually tested for viscosity as specified in 4.5.3.4.1 or, alternatively, as specified in 4.5.3.4.2 (see 6.5).

4.5.3.4.1 Bostwick Consistometer method.

Instrument: Bostwick Consistometer
Catalog number: 23270-004
VWR Scientific Company
P.O. Box 7900
San Francisco, CA 94120

or

Catalog Number: 15-347-50
Fisher Scientific
585 Alpha Drive
Pittsburgh, PA 15238

Method:

a. Level the instrument.

b. Bring gravy to 100°F ± 1°F in a water bath in a covered container.

c. Stir gravy thoroughly before filling the Bostwick cavity.

d. Scrape gravy evenly across upper edge of cavity.

e. Release gravy and time gravy flow to the nearest 1 second and measure distance traveled to the nearest 0.1 cm.

If the Bostwick viscosity value of the gravy from any can in the sample is less than 10.5 cm per 10 seconds or greater than 23.0 cm per 10 seconds, it is a major defect and the lot shall be rejected.

4.5.3.4.2 Brookfield Viscosimeter method.

Instrument: Brookfield Viscosimeter (variable speed) equipped with a No. 3 spindle.

Brookfield Engineering Lab., Inc.
Stoughton, MA 02072

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Method:

a. Bring gravy to 80°F ± 1°F in a water bath in a covered container.

b. Stir gravy thoroughly and pour 500 mL of the gravy into a 600 mL beaker.

c. Using a No. 3 spindle, stir at a speed of 10.0 rpm and record dial reading.

d. Convert dial reading to centipoise. Record result to nearest 100 centipoise.

If the Brookfield viscosity value of the gravy from any can in the sample is less than 1,300 centipoise or greater than 3,550 centipoise, it is a major defect and the lot shall be rejected.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the U.S. Standards for Condition of Food Containers except that inspection for labeling shall be as specified in 4.5.4.1.

4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4), except that for self-adhering labels the following additional defects shall apply:

Major: Label torn or scratched so as to obliterate any of the markings.

Minor: Air bubbles under label.
Label not properly adhered to can (i.e., label raised or peeled back from edges or corners).

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D 3330.

4.5.5 Can closure examination. Can closure shall be examined visually and by teardowns in accordance with the can manufacturer's requirement and 21 CFR, Part 113, Subpart D or 9 CFR, Part 318, Subpart G, as applicable. Any nonconformance based on observation of can seam teardowns or on record of can seam teardowns is a major defect and shall be cause for rejection of any involved product.

4.5.6 Vacuum examination. Cans shall be allowed to cool to 75° ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in

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the center of the tray pack can. When examining a ribbed lid, only lay the straight edge between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance is a major defect and shall be cause for rejection of the lot.

4.5.7 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Arrangement or number of cans not as specified.

4.5.8 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIl-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A or C, as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of the food product, sealed, and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.1.1 When specified (see 6.1), the base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 98 pounds per base box. The body design shall be reinforced on the bottom with six longitudinal beads or ribs 0.050 ± 0.010 inch deep. The beads shall be 9 inches by 3/4 inch in length and width and shall be equidistant in the center panel of the bottom. The end

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design shall be reinforced with eight longitudinal beads 0.018 ± 0.004 inch deep. The beads shall be 11 inches by 3/4 inch in length and width and shall be equidistant in the center panel of the end. The center four beads on the end design shall be interrupted to provide a 5-inch by 5-inch flat center panel for labeling. The apex of all these beads shall be outward. The interior and exterior can coating requirements shall be as specified in 5.1.1.

5.1.2 Level C. The product shall be preserved as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B, or C, as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with an HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1) except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

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5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified, except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (e.g., est 38) or a three-letter code identifying the establishment
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

1/ The lot number shall be expressed as a four-digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1987, would be coded as 7078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four-digit Julian code. Following the four-digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

Alternatively, a preprinted self-adhering, 0.002-inch thick, clear polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined as specified in 4.5.4 after preparation of the product in accordance with heating instructions. Paper labels are not permitted. Cans shall show the following statements:

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TO HEAT: Submerge unopened can in boiling water. Simmer gently 40 - 45 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place. Place in a 350°F oven 35 - 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions of 2/3 cup each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, when tray pack cans are fabricated from 90 pounds per base box, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT
OR MORE THAN TWO HIGH IN STORAGE,
UNLESS PALLET RACKS ARE USED.

6. NOTES

6.1 Ordering data. Acquistion documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of preservation and packing required (see 5.1 and 5.2).
e. When tray pack cans are to be fabricated from 98 pounds per base box (see 5.1.1.1).
f. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

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6.3 First article inspection. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Blanching process. Blanching of the beef chunks (see 3.3.1) to approximately a 70 to 75 percent yield has been found adequate in complying with finished product requirements.

6.5 Alternative viscosity testing method. The contracting officer may authorize an alternative contractor-recommended method of viscosity testing if the alternative method is approved by the U.S. Army Natick Research, Development and Engineering Center.

6.6 Subject term (key word) listing.

Beef tips
Canned foods
Combat field feeding
Thermostabilized
Tray pack

6.7 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, TS
Navy - MC
DP - SS

Preparing activity:

Army - GL

Project No. 8940-0603

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-44230A	Beef Tips With Gravy, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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